

A/B Test Sample Size and Power, Worked Out by Hand

Why a five-point lift on the refund agent needs $\sim 1,250$ runs *per arm* — and why the smaller the lift, the more violently that number explodes.

What this document does. It takes the one A/B test from Module 4.5 — the *refund-agent* rollout where candidate **v1.5.0** is measured against baseline **v1.4.2** on a 50/50 split — fixes a baseline success rate ($p_1 = 0.70$) and a target ($p_2 = 0.75$), and then computes, from first principles, how many runs per arm you must collect to call that 5-point lift real at $\alpha = 0.05$ with 80% power. Nothing is hand-waved. Every number below is reproduced by the small Python program in Appendix A, so the arithmetic is exact and self-consistent. The module quotes a 7-day partial rollout at 12,340 runs per arm; we derive the *floor* that number has to clear, and then show how it balloons as the lift shrinks.

Where this sits. This is **Topic 1 of Module 4.5** (Versioning & A/B Testing) — the math behind the “Statistical Significance” panel (success 95.4% $\rightarrow$ 97.2%, $p = 0.0013$). Its companions: Module 4.2 (observability — the dashboards that supply $\hat{p}_1, \hat{p}_2$) and Module 3.6 (evals — where the success metric itself is defined).

Concepts covered, in order: the two-proportion test · the significance level α and the critical value $z_{\alpha/2}$ · power $1 - \beta$ and z_β · pooled vs. per-arm variance · the sample-size formula $N = (z_{\alpha/2} + z_\beta)^2 [p_1(1-p_1) + p_2(1-p_2)] / (p_2 - p_1)^2$ · the inverse-square blow-up as the lift shrinks · variance inflation from LLM run-to-run non-determinism · why power, not significance, is the binding constraint for slow-moving agents.

Internal learning material. Numbers are reproducible from Appendix A. v1.0

1 The setup: one lift, small enough to size by hand

An A/B test on an agent asks one yes/no question: *is the candidate’s success rate really higher than the baseline’s, or did we just get lucky on this batch of runs?* The honest answer needs a sample large enough that a real lift would show through the noise. “Large enough” is not a vibe — it is a number you can compute *before* you ship, from four inputs:

Quantity	Symbol	Value here
Baseline success rate (v1.4.2)	p_1	0.70
Target success rate (v1.5.0)	p_2	0.75
Lift we want to detect	$\delta = p_2 - p_1$	0.05 (5 pp)
Significance level (two-sided)	α	0.05
critical value	$z_{\alpha/2} = z_{0.025}$	1.96
Power	$1 - \beta$	0.80
power value	$z_\beta = z_{0.20}$	0.84

Notation. N is the number of runs *per arm* (so the total experiment is $2N$). α is the false-positive rate: the chance of declaring a win when the arms are truly equal. $1 - \beta$ is power: the

chance of *catching* a lift of size δ when it is genuinely there. $z_{\alpha/2}$ and z_β are the standard-normal quantiles for those two error budgets. We carry the variance to four decimals throughout.

Intuition

Significance (α) and power ($1 - \beta$) are two different fears. α guards against *shipping a dud* — calling a tie a win. Power guards against *missing a real improvement* — shrugging off a genuine 5-point lift as noise. A test can be airtight on one and useless on the other. The sample-size formula is just the price, in runs, of buying down *both* fears at once to the levels you chose. Choose stricter levels, pay more runs.

2 Step 1 — the test statistic and its two quantiles

The metric is a proportion: out of N runs in an arm, some fraction succeeded. Comparing two proportions, the natural statistic is how many standard errors apart the two observed rates are. A real lift of size δ has to clear two hurdles:

- it must land far enough out that a true tie would rarely fake it — that distance is $z_{\alpha/2}$ standard errors (we use $\alpha/2$ because the test is two-sided: the candidate could be better *or* worse);
- and the *true* effect must sit far enough past that line that we clear it most of the time — an extra z_β standard errors of cushion.

So the detectable distance, in standard errors, is the sum $z_{\alpha/2} + z_\beta$. With the standard textbook values:

$$z_{\alpha/2} = z_{0.025} = 1.96, \quad z_\beta = z_{0.20} = 0.84, \quad z_{\alpha/2} + z_\beta = 2.80.$$

Mini-example — this operation in isolation

These two numbers are just normal-curve lookups, and they are worth memorising for the common case. $z_{0.025} = 1.96$ is the point with 2.5% of the normal mass in the upper tail — the famous “1.96” behind every 95% confidence interval. $z_{0.20} = 0.84$ is the point with 20% in the upper tail, which is exactly $1 - \text{power} = \beta = 0.20$ for 80% power. Their sum, 2.80, gets *squared* in the formula: $(2.80)^2 = 7.84$. That 7.84 is the “cost multiplier” of demanding $\alpha = 0.05$ and 80% power simultaneously. Want 90% power instead? $z_{0.10} = 1.28$, sum = 3.24, squared = 10.5 — a third more runs, just for the extra cushion.

3 Step 2 — the variance, and the sample-size formula

A single run is a Bernoulli trial: success with probability p , and its variance is $p(1 - p)$. The variance of an arm’s *mean* success rate over N runs is $p(1 - p)/N$. Comparing two arms, the variances add, so the difference $\hat{p}_2 - \hat{p}_1$ has variance $[p_1(1 - p_1) + p_2(1 - p_2)]/N$. Setting the detectable distance δ equal to $(z_{\alpha/2} + z_\beta)$ standard errors of that difference and solving for N gives the per-arm sample size:

$$N = \frac{(z_{\alpha/2} + z_\beta)^2 [p_1(1 - p_1) + p_2(1 - p_2)]}{(p_2 - p_1)^2} \quad (1)$$

Read it as three factors: a fixed error-budget cost $(z_{\alpha/2} + z_{\beta})^2$, a noise term in the numerator (how variable the outcomes are), and a signal term in the denominator (the lift, *squared*). Now plug in our numbers, piece by piece.

The variance term:

$$p_1(1 - p_1) = 0.70 \times 0.30 = 0.2100, \quad p_2(1 - p_2) = 0.75 \times 0.25 = 0.1875,$$

$$p_1(1 - p_1) + p_2(1 - p_2) = 0.2100 + 0.1875 = 0.3975.$$

The signal term: $(p_2 - p_1)^2 = (0.05)^2 = 0.0025$.

Assemble:

$$N = \frac{7.84 \times 0.3975}{0.0025} = \frac{3.1164}{0.0025} = 1246.56 \rightarrow \boxed{1247 \text{ runs per arm.}}$$

We always round *up* — a fractional run does not exist, and rounding down would quietly under-power the test. Sample check: $7.84 \times 0.3975 = 3.1164$; $3.1164/0.0025 = 1246.56$; $\lceil 1246.56 \rceil = 1247$. ✓ The total experiment is $2N = 2,494$ runs. The module's 12,340 per arm sits comfortably above this floor — which is exactly why its panel reaches $p = 0.0013$ rather than a marginal $p = 0.04$.

Intuition

Notice the asymmetry of the formula: the lift enters *squared* in the denominator, but the variance enters only linearly in the numerator. That single fact — signal squared, noise linear — is the whole reason small lifts are so brutally expensive to measure. Halving the lift you want to detect does not double the sample; it *quadruples* it.

4 Step 3 — the blow-up: N as the lift shrinks

Hold $p_1 = 0.70$ and α, β fixed; only move the target p_2 closer to baseline. Because δ sits squared in the denominator, N scales like $1/\delta^2$ — the steepest cliff in applied statistics.

Target p_2	Lift δ	numerator	N / arm	total $2N$
0.80	10 pp	2.9008	291	582
0.78	8 pp	2.9917	468	936
0.75	5 pp	3.1164	1,247	2,494
0.73	3 pp	3.1917	3,547	7,094
0.72	2 pp	3.2269	8,068	16,136
0.71	1 pp	3.2607	32,607	65,214

The numerator barely moves (the variance term is nearly flat near $p = 0.7$); the explosion is almost entirely the $1/\delta^2$ in the denominator. From a 5-point lift to a 2-point lift, δ shrank by $2.5\times$ but N grew $\sim 6.5\times$ ($1247 \rightarrow 8068$) — close to the $2.5^2 = 6.25$ the inverse-square law predicts. Chasing a 1-point lift demands 32,607 runs per arm: nearly three times the module's entire 7-day haul, *per arm*.

Sample check, $p_2 = 0.72$: variance = $0.21 + 0.72 \times 0.28 = 0.21 + 0.2016 = 0.4116$; numerator = $7.84 \times 0.4116 = 3.2269$; $N = 3.2269/(0.02)^2 = 3.2269/0.0004 = 8067.5 \rightarrow 8068$. ✓

Watch out

The same $1/\delta^2$ cliff sets a *detection floor* you may never reach. At the module’s throughput ($\approx 1,763$ runs/arm/day, from 12,340 over 7 days), a 5-point lift needs $1247/1763 \approx 0.7$ days — trivial. But a 1-point lift needs $32,607/1763 \approx 18.5$ days of clean, stationary traffic on *each* arm. Long before then your eval set drifts, the model provider ships a silent update, or seasonality moves the baseline — and the experiment is invalid. Some lifts are simply below your measurement floor; shipping them is then a judgment call, not a statistical one.

5 Step 4 — the agent twist: variance inflation from non-determinism

Equation (1) assumes each run is one clean Bernoulli draw. Agents break that assumption. As Module 4.5 warns, the *same* prompt at the *same* temperature gives different runs — so the success indicator carries run-to-run output noise *on top of* the underlying $p(1-p)$. The effect is to inflate the per-arm variance by a factor $\phi > 1$, which flows straight through the numerator:

$$N_{\text{eff}} = \phi \cdot N \quad \implies \quad \text{any inflation in variance is a proportional inflation in required sample.}$$

Inflation ϕ	Source	N_{eff} / arm	total
1.0	idealised, deterministic outcome	1,247	2,494
1.5	moderate run-to-run variability	1,871	3,742
2.0	high (chatty, multi-step agent)	2,494	4,988

The cure the module prescribes — *multi-seed baselines* — is exactly an attack on ϕ . Averaging each evaluation case over k seeds before scoring shrinks the run-to-run component of the variance (roughly by $1/k$ for that component), pulling ϕ back toward 1 and the required N back down. You pay for the variance either in wider arms or in repeated seeds; non-determinism does not let you pay nothing.

6 Step 5 — why power, not significance, is the binding constraint

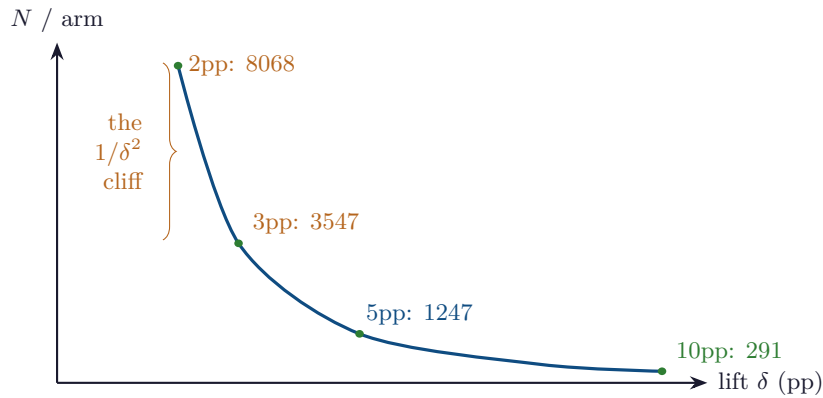
For a fast-moving consumer product, lifts are large and frequent, so α is the thing you sweat. For agents the opposite holds, and the formula says why. A matured agent improves *slowly*: each prompt revision buys a point or two of success, not ten. Small δ pushes you up the right tail of the $1/\delta^2$ curve, where the required N for adequate *power* dwarfs anything α asks for.

- Hitting $\alpha = 0.05$ is cheap: it only asks that the observed gap clear 1.96 standard errors. With enough runs that is almost automatic.
- Hitting 80% *power* on a 2-point lift costs 8,068 runs/arm — and on a 1-point lift, 32,607. *That* is what gates the rollout.
- So the practical question is rarely “is $p < 0.05$?” It is “do I even have the traffic to detect the lift I expect, before the world moves under me?”

Intuition

Under-powered agent A/Bs are the silent killer. A test with too few runs does not announce its weakness — it just returns “not significant” and you conclude the change did nothing, when in truth you simply could not see it. Repeat that across a quarter of prompt revisions and you convince yourself the agent has plateaued, while real gains slip past unmeasured. Sizing for power *first* — before you look at a single p-value — is the discipline that keeps an agent improving.

7 The sizing, drawn



The curve *is* Equation (1) with p_1 fixed: nearly flat where lifts are large and cheap to measure, then bending sharply upward as $\delta \rightarrow 0$. The green dot (10 pp) is a same-day test; the brown dots (3 and 2 pp) are the week-or-more tests that strain a real rollout. Where your expected lift lands on this curve decides whether the experiment is feasible at all.

8 Sample-size cheat-sheet

Per-arm sample size	$N = \frac{(z_{\alpha/2} + z_{\beta})^2 [p_1(1 - p_1) + p_2(1 - p_2)]}{(p_2 - p_1)^2}$
Total experiment	$2N$
Error-budget multiplier ($\alpha=.05$, power .80)	$(1.96 + 0.84)^2 = 7.84$
Variance term (per run)	$p_1(1 - p_1) + p_2(1 - p_2)$
Signal term	$(p_2 - p_1)^2 = \delta^2$
Scaling law in the lift	$N \propto 1/\delta^2$
LLM non-determinism	$N_{\text{eff}} = \phi N, \quad \phi > 1$
Multi-seed remedy	averaging k seeds shrinks the run-to-run part of ϕ

9 An honest word about these numbers

The rates here ($p_1 = 0.70$, $p_2 = 0.75$) are illustrative round numbers, chosen so the variance arithmetic is followable by hand — the module’s real figures sit higher (95.4% $\rightarrow$ 97.2%), where

$p(1 - p)$ is smaller and the required N is correspondingly lower for the same absolute lift. The quantiles $z_{0.025} = 1.96$ and $z_{0.20} = 0.84$ are standard rounded values; a stats package carries more digits and will nudge N by a run or two. The variance-inflation factor ϕ is a modelling choice, not a measured constant — you estimate it from your own multi-seed runs. What is *not* illustrative is the structure: N grows with the *square* of the error-budget quantiles and falls with the *square* of the lift, full stop. Change p_1, p_2, α, β and the numbers move; the $1/\delta^2$ shape — and the lesson that small agent lifts are expensive to prove — does not.

Appendix A — Reference implementation

Every number in this document is produced by the program below. Run it to reproduce the worked sample size, the lift-vs- N table, and the variance-inflation table; change p_1, p_2, z_a , or z_b to size your own experiment.

```

1  # ab_sample_size.py -- reproduces every number in "A/B Test Sample Size and Power".
2  import math
3
4  z_a = 1.96 # z_{alpha/2} = z_{0.025}, two-sided alpha=0.05
5  z_b = 0.84 # z_{beta} = z_{0.20}, power = 0.80
6
7  def n_per_arm(p1, p2):
8      # Eq. (1): two-proportion sample size, runs PER ARM.
9      num = (z_a + z_b)**2 * (p1*(1-p1) + p2*(1-p2))
10     den = (p2 - p1)**2
11     return num / den
12
13 # --- worked example: p1=0.70 -> p2=0.75 (5pp lift) ---
14 p1, p2 = 0.70, 0.75
15 var = p1*(1-p1) + p2*(1-p2)
16 print(f"(z_a+z_b)^2 = {(z_a+z_b)**2:.2f}")
17 print(f"variance term = {var:.4f}")
18 print(f"numerator = {(z_a+z_b)**2*var:.4f}")
19 N = n_per_arm(p1, p2)
20 print(f"N raw = {N:.2f} -> N/arm = {math.ceil(N)} total = {2*math.ceil(N)}")
21
22 # --- blow-up: N as the lift shrinks (p1 fixed at 0.70) ---
23 print("--- lift table ---")
24 for p2t in (0.80, 0.78, 0.75, 0.73, 0.72, 0.71):
25     num = (z_a + z_b)**2 * (p1*(1-p1) + p2t*(1-p2t))
26     N = math.ceil(n_per_arm(p1, p2t))
27     print(f"p2={p2t:.2f} lift={(p2t-p1)*100:4.0f}pp num={num:.4f} "
28           f"N/arm={N:6d} total={2*N}")
29
30 # --- variance inflation from LLM non-determinism ---
31 print("--- inflation table ---")
32 N0 = math.ceil(n_per_arm(0.70, 0.75))
33 for phi in (1.0, 1.5, 2.0):
34     Ne = math.ceil(N0 * phi)
35     print(f"phi={phi:.1f} N_eff/arm={Ne} total={2*Ne}")
36
37 # Expected output:
38 # (z_a+z_b)^2 = 7.84
39 # variance term = 0.3975
40 # numerator = 3.1164
41 # N raw = 1246.56 -> N/arm = 1247 total = 2494
42 # --- lift table ---
43 # p2=0.80 lift= 10pp num=2.9008 N/arm= 291 total=582
44 # p2=0.78 lift= 8pp num=2.9917 N/arm= 468 total=936
45 # p2=0.75 lift= 5pp num=3.1164 N/arm= 1247 total=2494
46 # p2=0.73 lift= 3pp num=3.1917 N/arm= 3547 total=7094

```

```
47 # p2=0.72 lift= 2pp num=3.2269 N/arm= 8068 total=16136
48 # p2=0.71 lift= 1pp num=3.2607 N/arm= 32607 total=65214
49 # --- inflation table ---
50 # phi=1.0 N_eff/arm=1247 total=2494
51 # phi=1.5 N_eff/arm=1871 total=3742
52 # phi=2.0 N_eff/arm=2494 total=4988
```

— *end of document* —